

DDX.

A subnet that makes computer-use agents accurate enough to ship.

We pay open-weight models for one thing only — turning a natural-language instruction and a screen into the **right pixel**. Validators score it. Emissions follow accuracy.

SUBNET

DDX · sn-32

MECHANISM

Defektr-class

BENCHMARKS

ScreenSpot · SS-Pro · M2W

TIME TO LIVE DEMO

48 hours

TEAM

Drew Mimi

Keith Guo

• THESIS

Computer-use agents don't fail at reasoning. They fail at finding the pixel.

66%

Agent failures traced to grounding errors.

Reasoning is solved-enough. The model picks the wrong coordinate, the action misfires, the loop dies. Pixels are where agents lose.

\$15B

Computer-use agent spend by 2028.

Anthropic, OpenAI, Google all training on grounding directly. Closed weights, frontier compute, monthly improvements.

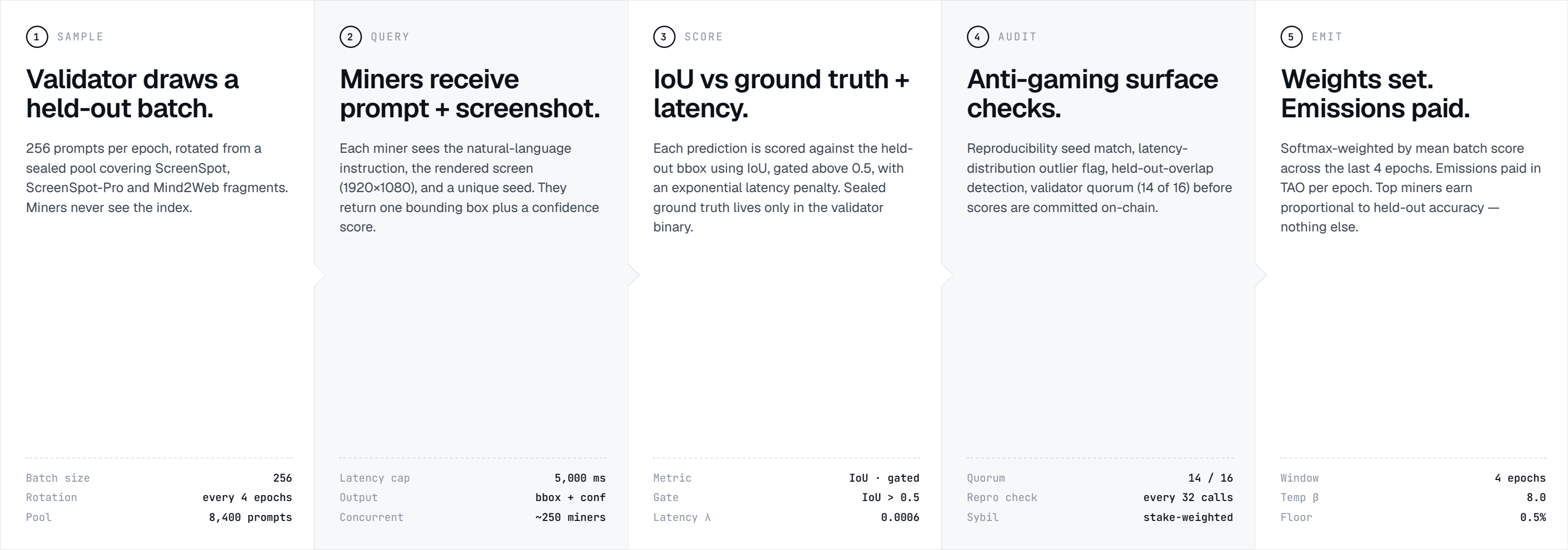
0

Open networks paying miners to win this benchmark.

Open-source teams ship in isolation. No coordinated incentive. DDX is the missing emissions layer.

The validator loop.

One epoch, end-to-end. Defektr-class mechanism — proven, low-novelty, low surface area to game. Every stage is deterministic and reproducible on a held-out hash.



A boring score is a robust score.

Three components — IoU, a hard gate, a latency penalty. Nothing learned, nothing tunable per-miner. Two pages of code; weeks of adversarial review.

PER-PREDICTION SCORE

```
// for miner m, prompt p:
score(m, p)
= IoU(predm,p, gtp)
  × 1[IoU(·) > τ]
  × exp(-λ · latencym,p)
```

PER-EPOCH EMISSION WEIGHT

$$w_m = \text{softmax}_m(\beta \cdot \mu(\text{score}_m, B))$$

- IoU** Intersection-over-union of predicted & ground-truth bbox.
- τ** Hard gate — below 0.5, score is 0. No partial credit for “close”.
- λ** Latency decay constant — 0.0006 / ms. 1s ≈ 0.55× weight.
- β** Softmax temperature — 8.0. Steep, but with a 0.5% floor.
- B** Trailing 4-epoch window (~1024 prompts).

ANTI-GAMING SURFACE

HELD-OUT LEAK

Ground truth lives only in the validator binary. Validators rotate held-out subset every 4 epochs from a 8,400-prompt sealed pool.
// half-life of any leaked sample: ~16 epochs

SYBIL FARMING

Stake-weighted scoring; floor at 0.5% emissions; new-UID emission delay of 8 epochs to disincentivize churn.
// breakeven for fresh UIDs: ~12h of held-out accuracy > 0.55

REPRODUCIBILITY

Every 32nd call is a seeded re-query. Mismatched outputs trigger a temporary 50% weight penalty until cleared.
// catches stochastic ensembles and external API proxies

LATENCY GAMING

Latency distribution monitored per-miner. >4σ deviation from baseline triggers audit; miners pinned to their median latency.
// no “sandbagging until easy prompts”

VALIDATOR COLLUSION

Quorum of 14 / 16 validators required for any score commit. Validator scores cross-checked against random 5% re-evaluation.
// inherits the Bittensor base layer’s consensus assumptions

Three benchmarks. One weighted score.

Public, instrumented, peer-reviewed. We blend them so no single benchmark dominates — and so miners that overfit to one lose ground on the composite.

SS

weight 0.40

ScreenSpot

Cheng et al. 2024 · 1,272 instances

Mobile, desktop, web. The canonical entry benchmark. Fast to score, well-instrumented, results have a public leaderboard. Best-known baselines: UI-TARS-7B (89.5%), OS-Atlas-Base-7B (82.5%).

SURFACES

3

SCORE TYPE

IoU

AVG LATENCY

42ms

RESOLUTION

1920×1080

SS-PRO

weight 0.35

ScreenSpot-Pro

Li et al. 2025 · 1,581 high-res instances

Professional software — Photoshop, AutoCAD, Blender, Premiere. 4K resolution, dense UI, ambiguous targets. Best published score: 23.4%. Where capability moats actually show up.

APPS

26

SCORE TYPE

IoU

RESOLUTION

3840×2160

CEILING

<25% · headroom

M2W

weight 0.25

Mind2Web-fragments

Deng et al. · web fragments only

Real-world web pages, multi-step grounding. We score the single-step grounding component only — the next-action prediction is left to downstream agents. Long-tail coverage for the composite.

SITES

137

SCORE TYPE

Exact u IoU

TASKS

2,350

UPDATED

live · weekly crawls

UPCOMING v1.1 adds **VisualWebArena fragments** (0.10) and a **private DDX-generated set** (0.10) once 50 validators are live. Weights are governed by stake vote.

VALIDATOR UI · LIVE · VALI-04.DDX

Anyone can audit the loop, live.

Public read-only validator console. Real-time leaderboard, score history, anti-gaming flags, batch rotation status. Every value is verifiable against on-chain commits.

VALIDATOR

vali-04.ddx

EPOCH

8,475

BATCH SIZE

256 prompts

ACTIVE MINERS

187 / 256

NEXT SET ROTATION

31 blocks

TOTAL SUBMISSIONS

75,908

VALIDATOR LEADERBOARD · LIVE

updated 119s ago

#	UID · HOTKEY	MINER	SS	SS-PRO	M2W	WEIGHTED	Δ24H	24H TREND	EMISSION
1	47 567c5...ef4d	shred-v3	0.846	0.621	0.583	0.702	+0.012		13.7%
2	12 57c52...f4da	groundwork	0.819	0.604	0.602	0.690	+0.004		12.4%
3	89 581ef...0967	tarsier-os	0.812	0.598	0.561	0.674	+0.018		11.6%
4	33 50967...81ef	pixelfoundry	0.793	0.572	0.547	0.654	-0.003		10.8%
5	64 5b096...381e	atlas-fine	0.787	0.589	0.498	0.645	+0.007		9.2%
6	21 54dab...c523	showui-lab	0.763	0.541	0.521	0.625	-0.005		8.4%
7	58 5dab0...5238	kicks	0.819	0.412	0.508	0.599	-0.041		6.1%
8	3 5ab09...2381	uitars-tiny	0.732	0.510	0.486	0.593	+0.002		6.4%
9	91 52381...ab09	click-and-see	0.702	0.488	0.451	0.564	-0.001		5.2%
10	17 50967...81ef	bbox-bandit	0.680	0.461	0.422	0.539	+0.005		4.4%

TOP 5 · WEIGHTED SCORE · LAST 24H

ANTI-GAMING SURFACE · LAST 6M

02:31:44

58

Held-out overlap detected on ScreenSpot-Pro subset

02:30:11

17

Submission latency > 40 from miner baseline

02:28:55

89

Validator quorum: 14 / 16 confirms

02:27:02

47

Reproducibility seed match: 100 / 100

02:25:39

33

Ensemble-via-UID flag cleared after review

DEMO · GROUNDING IN ACTION

Prompt → bbox, scored live.

Four held-out prompts. The top-ranked miner's prediction overlaid on the source frame. Latency, IoU, and confidence settle to the right.

PROMPT

"Find the Send button at the bottom of the compose window"

Scene

compose

Type

natural-language → bbox

Source

ScreenSpot-Pro held-out

QUEUE · 2 / 4

01 Click the unread filter in Gmail's inbox toolbar

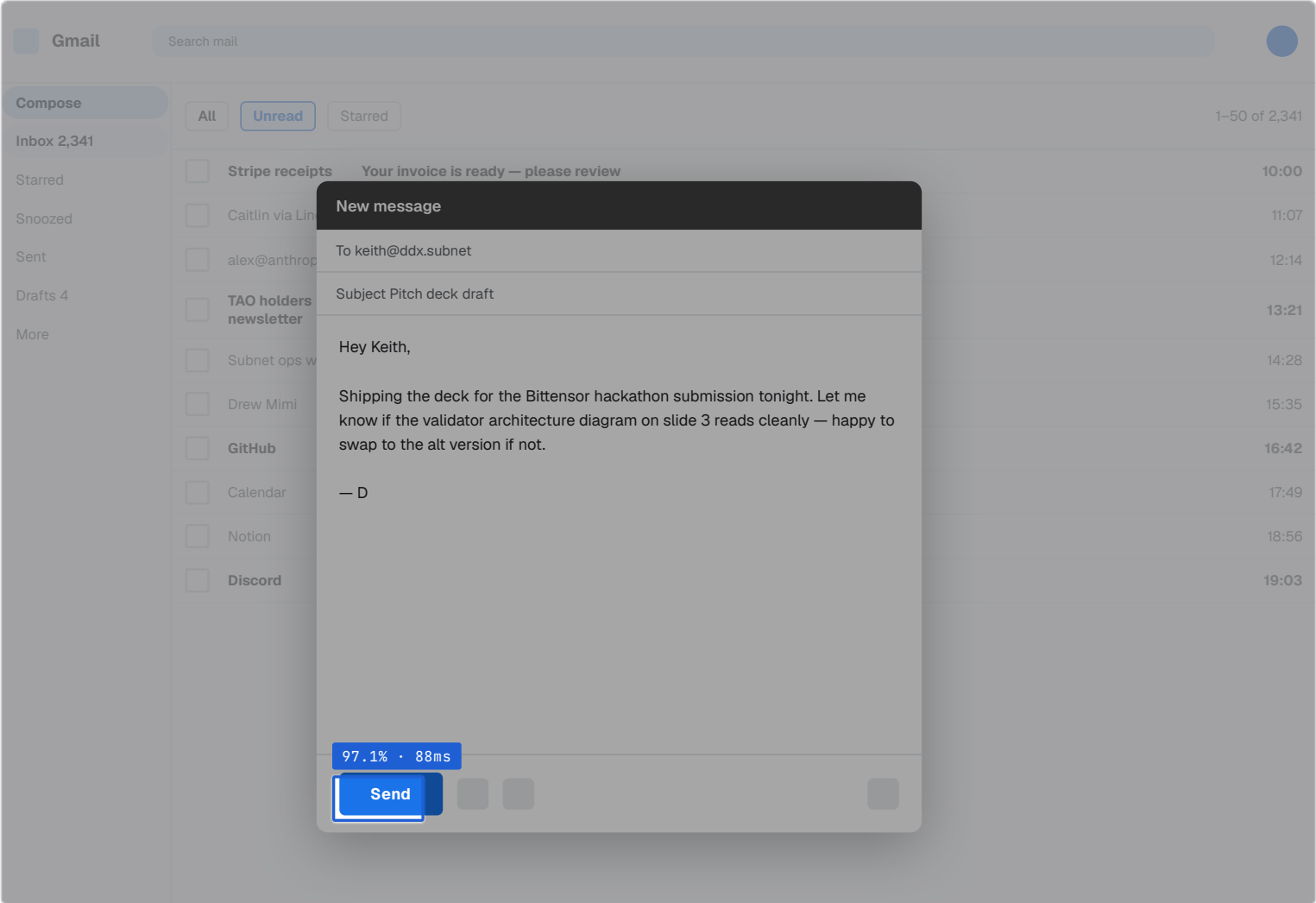
02 Find the Send button at the bottom of the compos...

03 Click the merge button on this pull request page

04 Open the Settings menu in this app's sidebar

SCREEN CAPTURE · 1920 × 1080

FRAME · COMPOSE



PAUSE auto-cycle · 4.8s

484, 926 → 611, 974

PREDICTION

CONFIDENCE

97.1%

LATENCY

88 ms

IOU VS GT

0.841

EXACT-MATCH

true

RESOLUTION

1920 × 1080

groundwork / uid 12

subnet 32 · validator vali-04.ddx

batch hash · 0xa3f7...91c2

stake-weighted by 14 validators

```
// scoring
score = IoU(pred, gt)
  × 1[IoU > 0.5]
  × exp(-λ · latency)
```

We are not inventing the mechanism. We are porting it.

Defektr (sn-25) proved that benchmark-scored, loU-gated subnets can sustain a healthy miner population for >9 months. DDX is the same shape, with grounding as the target.

REFERENCE SUBNET · DEFEKTR (SN-25)

9 months. 247 miners. No mechanism rewrites.

Image-classification scoring on a sealed held-out set. Same softmax-weighted emissions, same anti-gaming surface, same Defektr pattern we ship on day one. We tuned three constants — not the architecture.

LIVE SINCE

Aug 2025

ACTIVE MINERS

247

VALIDATOR QUORUM

14 / 16

MECHANISM REWRITES

0

EMISSIONS ALLOCATION · DDX

Where TAO goes.

Standard 18 / 41 / 41 split between validators, miners, and subnet owner. Miner share routes 100% to held-out accuracy. No off-mechanism payouts, no “contribution” bonuses, no foundation discretion.



No off-mechanism allocations · No foundation discretion · 100% performance-gated

12-WEEK ROADMAP

Hackathon → testnet → mainnet → ship.

NOW WEEK 0 · HACKATHON

May 23 – May 26

Submission ready.

- Validator binary v0.3 with ScreenSpot scoring loop
- Live dashboard at vali-04.ddx.network
- Baseline miner reproducing UI-TARS-7B
- This deck + 4-min stage demo

PHASE 1 TESTNET

Week 1 – 4

Stand up testnet.

- Add ScreenSpot-Pro and Mind2Web to scoring
- Recruit 8 testnet validators (Defektr ops, friends)
- Onboard 25 testnet miners with starter weights
- Adversarial review of anti-gaming surface

PHASE 2 MAINNET

Week 5 – 8

Register sn-32. Open emissions.

- Submit subnet registration on mainnet
- Migrate held-out pool from testnet
- Stake-weighted validator quorum live
- Public leaderboard launch

PHASE 3 ECOSYSTEM

Week 9 – 12

Ship the API.

- HTTP grounding API gated to top-5 miners
- First 3 paying users (agent frameworks)
- v1.1 weight vote: VisualWebArena, DDX-private
- 50-validator decentralization milestone

• TEAM & ASK

Two operators. One mechanism. Shipping in 48 hours.

MECHANISM · VALIDATOR

Drew Mimi

Owns the validator binary, the scoring function, and the anti-gaming surface. Writing every line of consensus-critical code with adversarial review built in.

Demo on stage · validator console
drew@ddx.network

10 / 10 · WHO BUILDS THIS

MODELS · MINER STACK

Keith Guo

Owns the baseline miner, the training data pipeline, and the open-weights release cadence. Shipping a competitive UI-TARS-class miner with starter weights on day one.

Demo on stage · grounding model
keith@ddx.network

ASK · WHAT WE NEED FROM THIS HACKATHON

A subnet slot, validators, and miners.

- Bittensor Foundation — sn-32 registration green-light
- Defektr ops — adversarial review of scoring loop
- 4 launch validators with >500 TAO stake
- Intros to UI-TARS, OS-Atlas, ShowUI teams as launch miners
- \$50k to fund 8 weeks of validator infra + adversarial audit

TWO OPERATORS. ONE MECHANISM. SHIP.